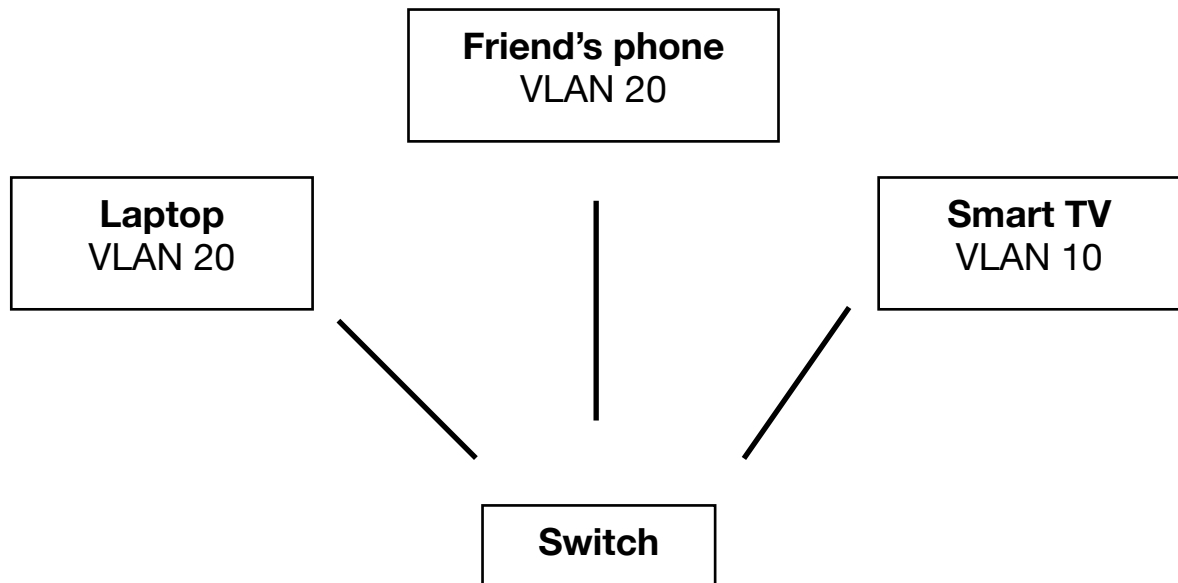


Module 1: Network + Network Fundamentals and building networks

Switching and VLANs

Task 1:



Task 2:

Create VLAN 10 & 20, assign PCs, verify isolation

```
Switch> enable
Switch# configure terminal
```

Create the VLANs

```
Switch(config)# vlan 10
Switch(config-vlan)# name Streaming
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name Personal
Switch(config-vlan)# exit
```

Assign PC1, PC2, PC3 to VLAN 10

```
Switch(config)# interface range fastEthernet 0/1-3
Switch(config-if-range)# switchport mode access
```

```
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
```

Assign PC4, PC5 to VLAN 20

```
Switch(config)# interface range fastEthernet 0/4-5
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit
```

```
Switch(config)# exit
Switch# show vlan brief      ! verify VLAN-to-port assignment
```

Task 3:

Moving a PC from VLAN 10 to VLAN 20

Before (PC3 is in VLAN 10, on port fa0/3):

```
Switch(config)# interface fastEthernet 0/3
Switch(config-if)# switchport access vlan 10Switch(config)# interface fastEthernet
0/3
Switch(config-if)# switchport access vlan 10
```

After (moving PC3 to VLAN 20):

```
Switch(config)# interface fastEthernet 0/3
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
Switch(config)# exit
Switch# show vlan brief      ! confirm fa0/3 now shows under VLAN 20
```

Test connectivity:

- Before the change: PC3 could ping other VLAN 10 devices, but not VLAN 20 devices.
- After the change: ping a VLAN 20 device from PC3 — it should now succeed, while pinging a VLAN 10 device should now fail, confirming the move took effect.

Task 4:

Automate assigning 10 ports to VLAN 30

```
Switch(config)# vlan 30
Switch(config-vlan)# name Guest
Switch(config-vlan)# exit
```

```
Switch(config)# interface range fastEthernet 0/11-20
Switch(config-if-range)# switchport mode access
```

```
Switch(config-if-range)# switchport access vlan 30  
Switch(config-if-range)# exit
```

```
Switch(config)# exit  
Switch# show vlan brief
```